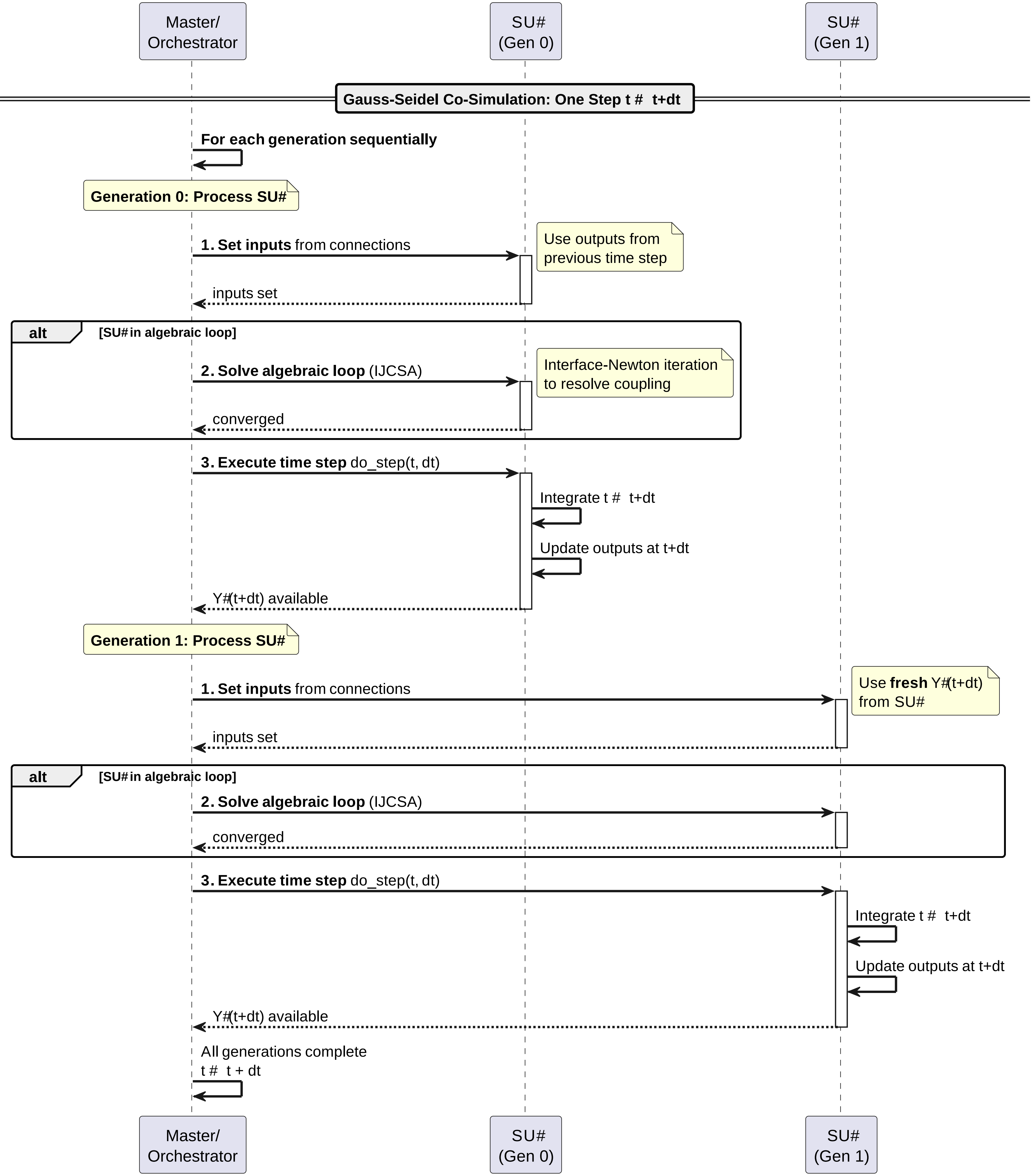

Master/ Orchestrator
SU# (Gen 0)
SU# (Gen 1)
Gauss-Seidel Co-Simulation: One Step t # t+dt
For each generation sequentially
Generation 0: Process SU#
1. Set inputs from connections
Use outputs from previous time step
inputs set
alt
[SU# in algebraic loop]
2. Solve algebraic loop (IJCSA)
Interface-Newton iteration to resolve coupling
converged
3. Execute time step do_step(t, dt)
Integrate t # t+dt
Update outputs at t+dt
Y#(t+dt) available
Generation 1: Process SU#
1. Set inputs from connections
Use fresh Y#(t+dt) from SU#
inputs set
alt
[SU# in algebraic loop]
2. Solve algebraic loop (IJCSA)
converged
3. Execute time step do_step(t, dt)
Integrate t # t+dt
Update outputs at t+dt
Y#(t+dt) available
All generations complete
t # t + dt
Master/ Orchestrator
SU# (Gen 0)
SU# (Gen 1)